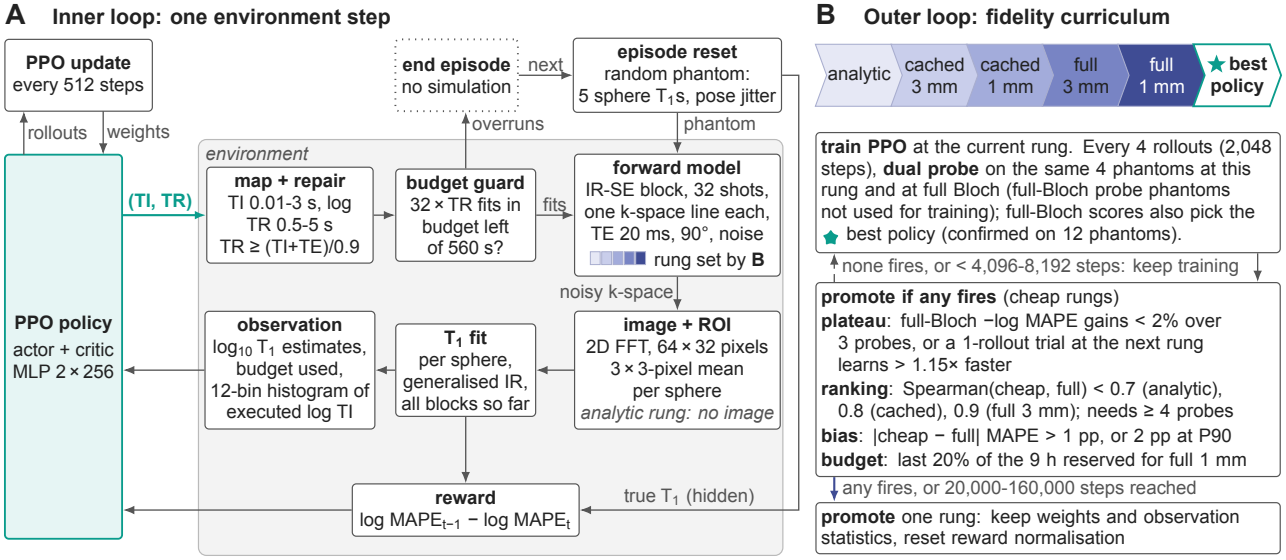

A
Inner loop: one environment step
PPO update
every 512 steps
rollouts
weights
PPO policy
actor + critic
MLP 2 × 256
(TI, TR)
environment
map + repair
TI 0.01-3 s, log
TR 0.5-5 s
TR ≥ (TI+TE)/0.9
budget guard
32 × TR fits in
budget left
of 560 s?
overruns
fits
end episode
no simulation
next
episode reset
random phantom:
5 sphere T1s, pose jitter
phantom
forward model
IR-SE block, 32 shots,
one k-space line each,
TE 20 ms, 90°, noise
rung set by B
noisy k-space
image + ROI
2D FFT, 64 × 32 pixels
3 × 3-pixel mean
per sphere
analytic rung: no image
T1 fit
per sphere,
generalised IR,
all blocks so far
observation
log10 T1 estimates,
budget used,
12-bin histogram of
executed log TI
reward
log MAPEt−1 − log MAPEt
true T1 (hidden)
B
Outer loop: fidelity curriculum
analytic
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm
best policy
train PPO at the current rung. Every 4 rollouts (2,048 steps), dual probe on the same 4 phantoms at this rung and at full Bloch (full-Bloch probe phantoms not used for training); full-Bloch scores also pick the best policy (confirmed on 12 phantoms).
none fires, or < 4,096-8,192 steps: keep training
promote if any fires (cheap rungs)
plateau: full-Bloch −log MAPE gains < 2% over 3 probes, or a 1-rollout trial at the next rung learns > 1.15× faster
ranking: Spearman(cheap, full) < 0.7 (analytic), 0.8 (cached), 0.9 (full 3 mm); needs ≥ 4 probes
bias: |cheap − full| MAPE > 1 pp, or 2 pp at P90
budget: last 20% of the 9 h reserved for full 1 mm
any fires, or 20,000-160,000 steps reached
promote one rung: keep weights and observation statistics, reset reward normalisation